

* Revision:

- ① Multiple LR: more than one feature.
- ② overfitting & underfitting:

g: if training error $\downarrow$ & testing $\uparrow$:

if $\uparrow$ error $\uparrow$ & testing $\uparrow$:

if $\uparrow$ & $\downarrow$:

very less data:

very simple model:

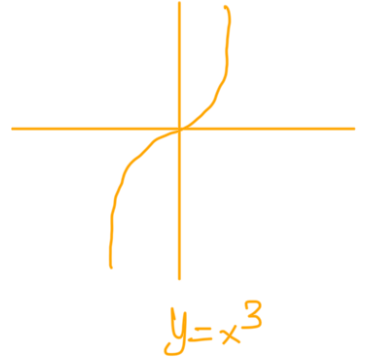
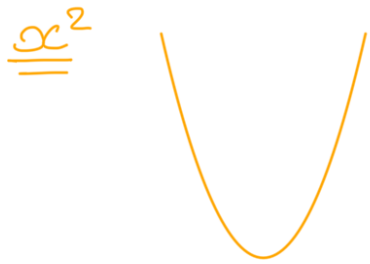
* today's class:

- ① Polynomial Regression:

Does we have always a linear / linear sort of data ? NO:



→ To Fit model into this pattern / shape we require to add some non linearity.



⇒ Now our equation becomes:

$$y = mx + b \Rightarrow y = m_1x + m_2x^2 + m_3x^3 + \dots + m_nx^n + b$$

⇒ Power of x : (degree)

↳ degree is hyperparameter
↓
We have to decide it.

$x^2 \rightarrow 2$ $y = m_1x + m_2x^2 + b$

$x^3 \rightarrow 3$ $y = m_1x + m_2x^2 + m_3x^3 + b$

$x^4 \rightarrow 4$ $y = m_1x + m_2x^2 + m_3x^3 + m_4x^4 + b$

⇒ How to decide degree(n):

Go to Visualisation tool. : As degree Increase the curve can bend more & more.

→ Stop before overfitting:

A degree for which we get lowest test Error.
(Lowest error on unseen data)

low degree : under fitting

high : over fitting

perfect degree btwn low & high

⇒ What is poly Reg.?

Polynomial Reg. is a special case of L.R. where features are transformed into polynomial terms to capture non linearity.

$$y = mx + b \xrightarrow[\text{Poly}]{d=3} y = m_1x + m_2x^2 + m_3x^3 + b$$

$x \rightarrow x_1$
$x^2 \rightarrow x_2$
$x^3 \rightarrow x_3$

$$y = m_1x_1 + m_2x_2 + m_3x_3 + b$$

Does it look like
MLR? YES

So we train it as MLR.

⇒ Go To Kaggle Note book.

1st: training

2nd: $d=2$ & $d=3$ Results

3rd: transformation of features

degree=2 og features: 6 $\xrightarrow[\text{poly}]{d=2}$ 28

$$y = m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4 + m_5x_5 + m_6x_6 + b$$

↪

$$y = m_1x_1 + m_2x_2 + \dots + m_{28}x_{28} + b$$

x_1	$\rightarrow$	x_1^2	$\rightarrow$	x_1x_2	$\rightarrow$	x_2x_3	$\rightarrow$	x_3x_4	$\rightarrow$	x_4x_5	$\rightarrow$	x_5x_6
x_2	$\rightarrow$	x_2^2	$\rightarrow$	x_1x_3	$\rightarrow$	x_2x_4	$\rightarrow$	x_3x_5	$\rightarrow$	x_4x_6		
x_3	$\rightarrow$	x_3^2	$\rightarrow$	x_1x_4	$\rightarrow$	x_2x_5	$\rightarrow$	x_3x_6				
x_4	$\rightarrow$	x_4^2	$\rightarrow$	x_1x_5	$\rightarrow$	x_2x_6						
x_5	$\rightarrow$	x_5^2	$\rightarrow$	x_1x_6								
x_6	$\rightarrow$	x_6^2										
$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$		$\downarrow$
6	+	6	+	5	+	4	+	3	+	2	+	1
												<u>$= 27 + 1$ bias.</u>

training of poly: transform features into poly. features.

↓
old + new features
↓
Train it as Linear Regression.

Note: Before training of model using poly features we apply feature scaling.

* Regression Metrics:

$$\underline{MAE}: \frac{1}{n} \sum |y - \hat{y}| \quad \underline{MSE}: \frac{\sum (y - \hat{y})^2}{n} \quad \underline{RMSE}: \sqrt{MSE}$$

Why $1/n$: (why mean/Avg Error ? why not total Error ?)

Ex: 5 data points $\xrightarrow{\text{same prediction}}$ 10 data points

- That's why it makes standard

G.D. Inconvenient.

② MSE: $\frac{1}{n} \sum (y - \hat{y})^2$

- we can use it as loss function with G.D.

$(1PA \rightarrow (1PA)^2)$

- not same unit.
- not Robust to outliers.

$(5 \rightarrow 25)$

	AE	SE
2 x	↓	↓ <u>x 4</u>
	10	100

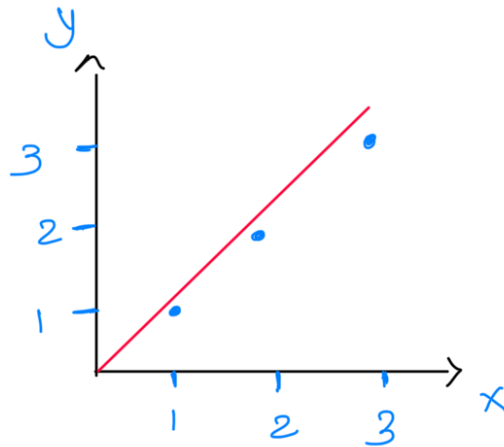
- very sensity to outliers.

③ RMSE: $\sqrt{MSE}$

AE SE
1PA $\rightarrow$ (1PA)² $\rightarrow$ $\sqrt{(1PA)^2} = 1PA$
Same unit

Not Robust to outliers.

④ R^2 score:



$\left\{ \begin{matrix} x & y \\ (1, 1) \\ (2, 2) \end{matrix} \right\} \quad \left\{ \begin{matrix} x & y \\ (3, 3) \end{matrix} \right\}$

Avg of $y = \frac{1+2+3}{3} = \boxed{2}$

x	$\hat{y}$	$\bar{y}$	$\hat{y} - \bar{y}$	$\bar{y} - \bar{y}$
1	1.2	2	0.2	1.1
2	2.2	2	0.2	0
3	3.3	2	0.3	-1.1
			0.7	2

→ R^2 basically compares our model against a mean prediction base-line.

$$R^2 = 1 - \frac{SSR}{SSM} \rightarrow \begin{matrix} \text{Error from our model} \\ \text{Error from mean line} \end{matrix}$$

SSR = Sum of Squared Error on Regression line
 SSM = " " " " " on mean line
 $\rightarrow SSR (y - \hat{y})^2$ [$\bar{y}$ = mean / Avg]
 $\rightarrow SSM (y - \bar{y})^2$

$$R^2 = 1 - \frac{SSR}{SSM}$$

Ae jitna kam. Hoga
 utano Hamara
 $\frac{SSR}{SSM}$ kam Hoga.

SSM ko ham
 kam nahi kar skate. It will be constant.

$$SSR \downarrow \rightarrow \frac{SSR}{SSM} \downarrow \rightarrow \left(1 - \frac{SSR}{SSM}\right) \uparrow \rightarrow R^2 \uparrow$$

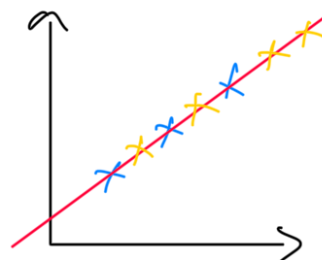
Perfect model : $SSR = 0$

$\downarrow$

$$\frac{SSR}{SSM} = 0$$

$\downarrow$

$$\boxed{R^2 = 1}$$



∴ $R^2 = 0$ $\rightarrow$ When $SSR = SSM$

$$1 - \frac{SSR}{SSM} = 1 - 1 = 0$$

$\rightarrow$ Our model is performing as
 same as mean line.

∴ $R^2 < 0$: $SSR > SSM$ $\rightarrow$ $1 - (>1)$: < 0 value
 R^2

$\rightarrow$ Our model is performing worse than
 simply predicting the mean.

$\Rightarrow$ Range of R^2 : $(-\infty, 1]$

⇒ definition 2: Proportion (percentage) of the total
variance in the target explained by
the model.

$R^2 = 0.8$: 80 % variance can be explained
by our model.